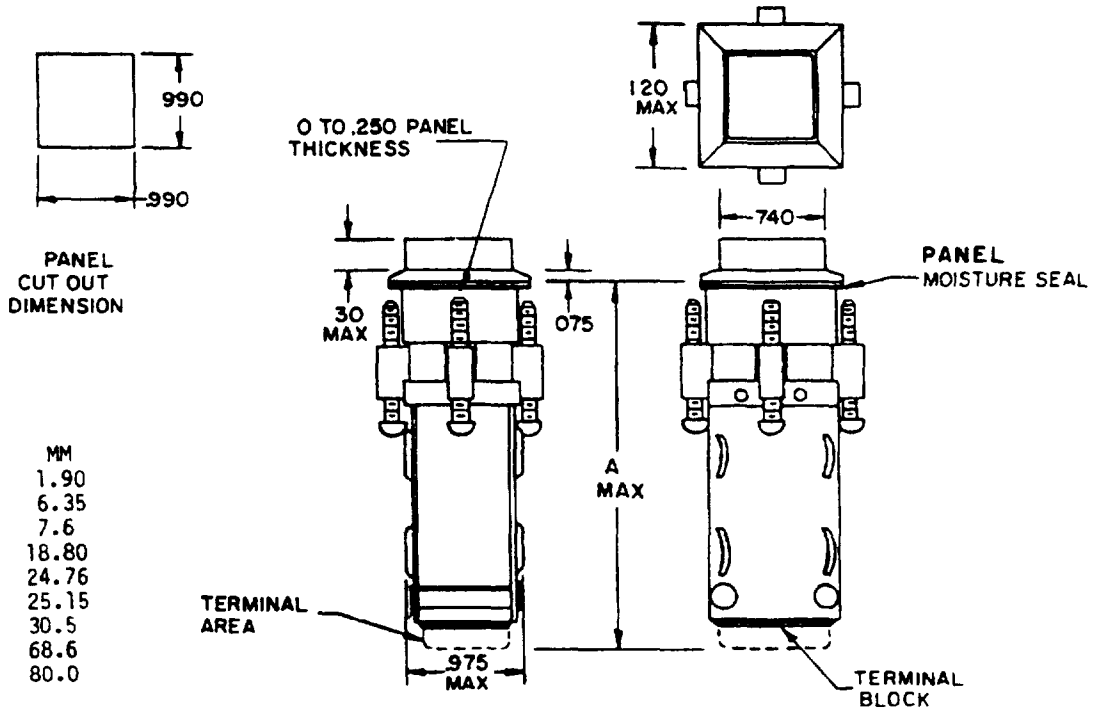
MILITARY SPECIFICATION SHEET

SWITCH, PUSH BUTTON, ILLUMINATED, 4-LAMP, SOLID MOUNT, RFI SHIELDED, MOISTURE SEALED, HIGH IMPACT SHOCK RESISTANT

This specification sheet is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the switches described herein shall consist of this specification and the latest issue of MIL-S-22885.



INCHES	MM
.075	1.90
.250	6.35
.30	7.6
.740	18.80
.975	24.76
.990	25.15
1.20	30.5
2.70	68.6
3.15	80.0

DIMENSION A

TYPE 1: 3.15 MAX
TYPE 3: 2.70 MAX

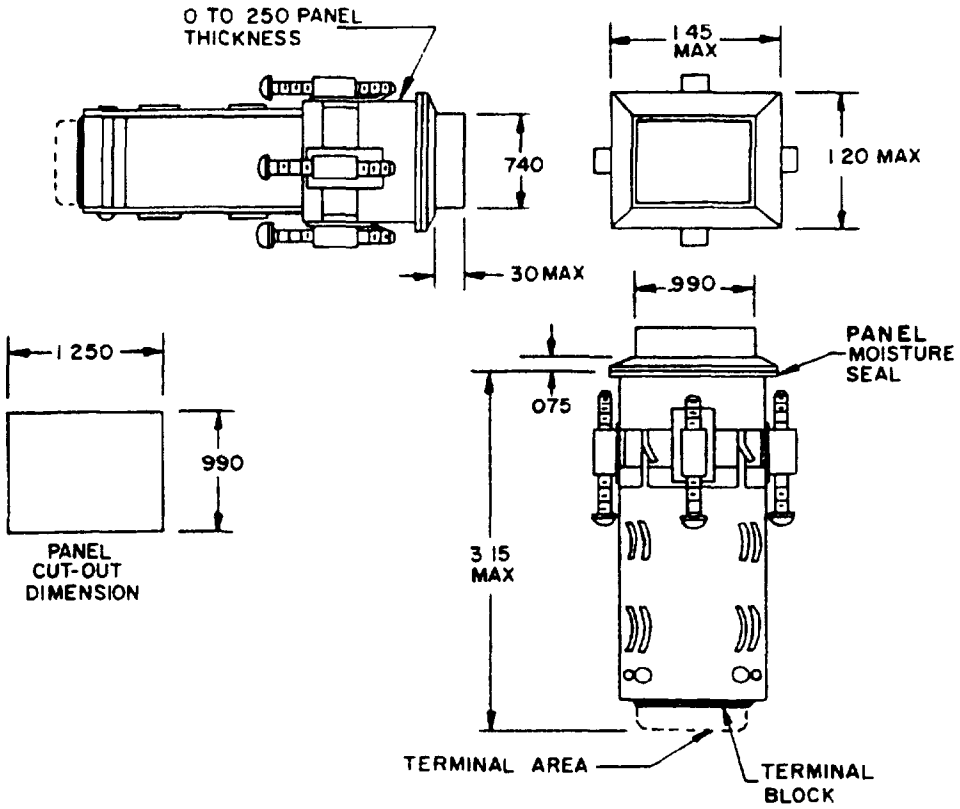
NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.
3. Unless otherwise specified, tolerances are ±.010 (.25 mm).
4. The design configuration is optional within the envelope dimensions shown.

FIGURE 1. Switch, 4-lamp, solid mount, type 1 and type 3.

Ⓔ denotes changes

FSC 5930

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T14:25Z
Check the source to verify that this is the current version before use.

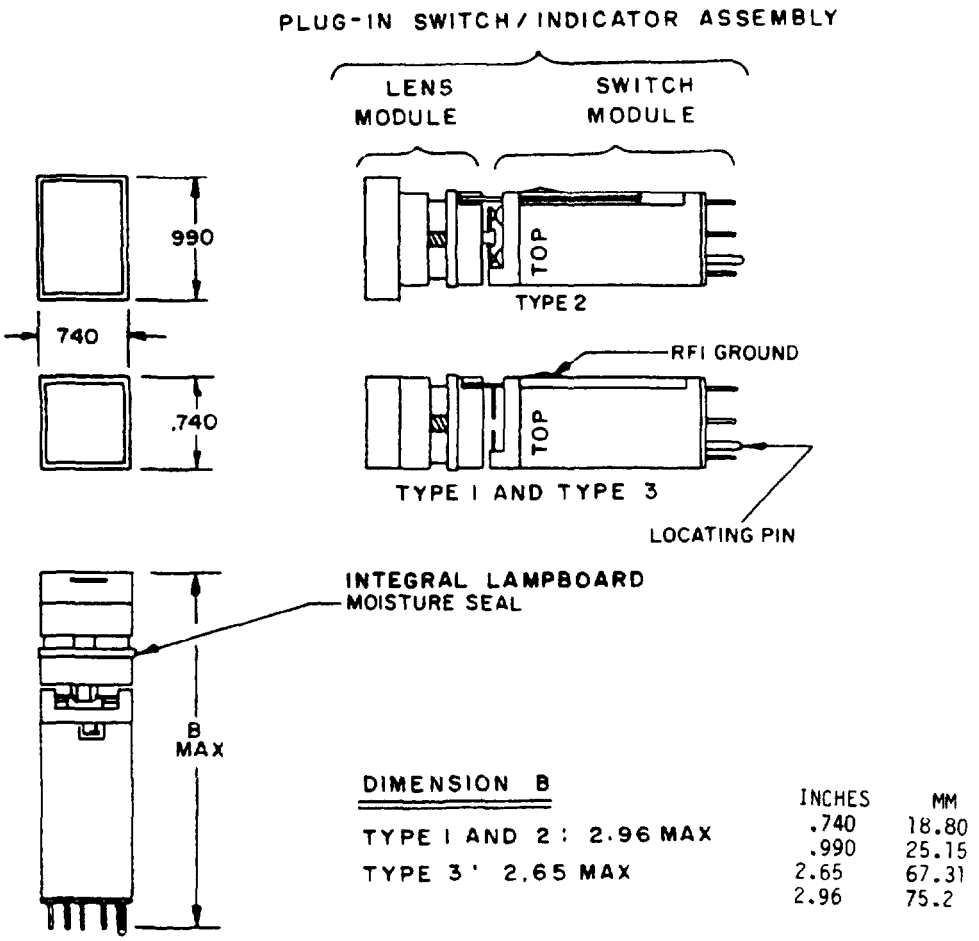


INCHES	MM
.075	1.90
.250	6.35
.30	7.6
.740	18.80
.990	25.15
1.20	30.5
1.250	32.75
1.45	36.8
3.15	80.0

NOTES
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.
3. Unless otherwise specified, tolerances are ±.010 (.25 mm).
4. The design configuration is optional within the envelope dimensions shown.

FIGURE 2. Switch, 4-lamp, solid mount, type 2.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T14:25Z
Check the source to verify that this is the current version before use.

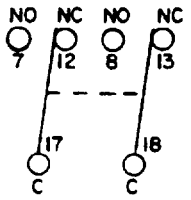


DIMENSION B	INCHES	MM
TYPE 1 AND 2: 2.96 MAX	.740	18.80
TYPE 3: 2.65 MAX	.990	25.15
	2.65	67.31
	2.96	75.2

NOTES.
1. Dimensions are in inches and shall be used for reference only.
2. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.
3. Unless otherwise specified, tolerances are ±.016 (.41 mm).

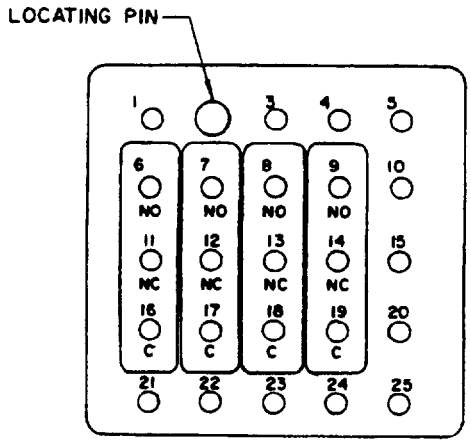
FIGURE 3. Plug-in switch/indicator assembly.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T14:25Z
Check the source to verify that this is the current version before use.

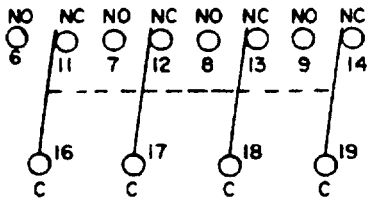


Switch Circuitry

2 PDT Momentary and Alternate Action



(Rear View - Terminal Block)
Switch Terminal Identification



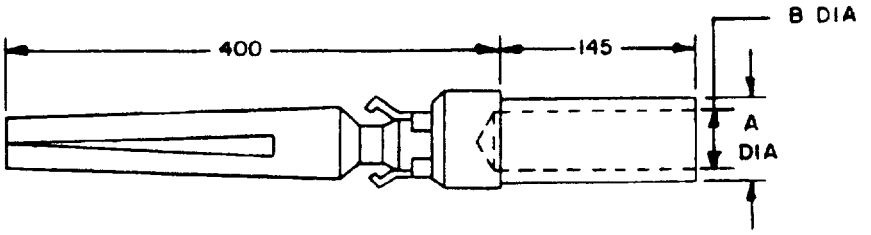
Switch Circuitry

4 PDT Momentary and Alternate Action

NOTE Terminals shall be permanently identified as shown.

FIGURE 4. Switch circuitry schematics.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T14:25Z
Check the source to verify that this is the current version before use.



Terminal part numbers and selection criteria.

P/N M22885/80- (package of 25 terminals)	"A" dia.	"B" dia.	Acceptable wire sizes & qty. (stranded or solid conductor) (see note 5)
50	.076	.054	#20(1), #22(1), #24(1) or #24(2)
51	.049	.029	#26(1) or #28(1)
52	.092	.072	#18(1), #20(2) or #22(2)

INCHES	MM
.029	.74
.049	1.24
.054	1.37
.072	1.83
.076	1.93
.092	2.34
.145	3.68
.400	10.16

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.
3. Unless otherwise specified, tolerances are ±.010 (.25 mm).
4. Design configuration is optional within envelope dimensions shown.
5. Wire size for stranded conductor is based on nearest AWG size.
6. Terminals are crimped onto the end of each wire using M22520/1-01 crimping tool and M22520/1-02 positioner in accordance with MIL-C-22520.

FIGURE 5. Terminal, crimp-type.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T14:25Z
Check the source to verify that this is the current version before use.

REQUIREMENTS

Design and construction

Dimensions and configuration: See figures 1 through 5.

Complete switch shall consist of

1 - Plug-in switch indicator assembly (see figure 3)

1 - Housing.

4 - Lamps (T-1 3/4 midget-flanged-base). Not included, order separately.

Plug-in switch/indicator assembly shall include

1 - Switch module

1 - Lens module Consists of lens assembly in accordance with MIL-S-22885/82, as specified (see table IV and V), lens retainer (M22885/82-50 or 58), and lamp capsule assembly with RFI screen and moisture seal. The lamps and lens assembly shall be capable of being installed or removed from the panel front without the use of special tools.

Housing shall include

1 - Terminal block.

1 - Mounting fastener.

Terminals See figure 5 for available part numbers. Not included, order separately.

Enclosure design: 2 (Dripproof).

Temperature characteristic

-55°C minimum to +71°C maximum.

+55°C maximum with lamps energized.

Vibration grade 2 (10-500 Hz).

Operation A (momentary), B (alternate), H (indicator only).

Color See table II and MIL-S-22885/82.

Display type W (white until illuminated, then background appears in color specified).

Material:

Plug-in terminations Gold plating .000050 inch minimum thickness per MIL-G-45204, type II, class I over copper plate per MIL-C-14550.

(E) Housing bezel Housing bezel exposed at front of the panel shall have gold irridite finish (chemical filmed in accordance with MIL-C-5541).

Weight

Type 1 3.0 ounces maximum.

Type 2 3.5 ounces maximum.

Type 3 2.5 ounces maximum.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T14:25Z
Check the source to verify that this is the current version before use.

Solderability. Not applicable.

Resistance to soldering heat: Not applicable.

(E) Operating characteristics.

Actuating force: 4 pounds maximum.

Shock.

Momentary action switches and indicator. Method I and method II (high impact). Displacement of lens module to re-lamp position permissible during method II high impact shock test.

Alternate action switches. Method I.

(E) Seal No leakage allowed through panel seal and beyond integral lamphoard seal (lens assembly not sealed)

Electrical endurance. Test at electrical ratings specified in table I.

Contact resistance Not applicable after electrical endurance, inductive load dc.

EMI/RFI shielding. Grounded lens screen shall be provided.

Luminance: See table II.

Marking. Part numbers for complete assemblies shall not appear on the items. The part numbers for individual plug-in switch/indicators and housing shall be marked on the item as required.

TABLE I. Electrical ratings.

Sea level		
Load	28 V dc	115 V ac 60 Hz
	(amperes)	(amperes)
Resistive	5.0	5.0
Inductive	2.5	2.5

TABLE II. Luminance.

Color	Minimum switch photometric brightness (footlamberts)
Green (G)	8
Yellow (Y)	150
Red (R)	40
Blue (B)	3
White (W)	55

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T14:25Z
Check the source to verify that this is the current version before use.

QUALIFICATION

Group submission. The following applies when the momentary action switches, types 1 and 2, are submitted for qualification at the same time.

Qualification inspection. See table III.

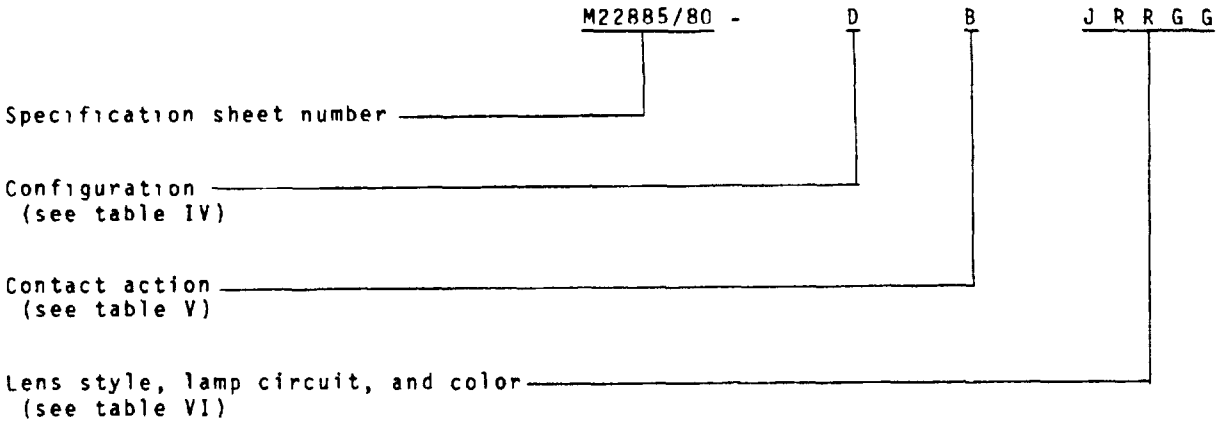
TABLE III. Qualification inspection.

Part number	Test	Extent of approval
M22885/80-ABCW 1/	Qualification inspection table of MIL-S-22885 2/	All part numbers
M22885/80-FDLRW	Groups I and II tests from qualification inspection table	

1/ Switches with red, green, yellow and blue color filters shall be tested, in addition to white, for group VIII tests.

2/ Two samples shall be submitted to method I shock and two samples to method II shock.

Part number. Part numbers shall be assigned as illustrated below. For acquisition of government spares, use configuration codes for plug-in switch/indicator assembly and housing only. Provisioning documentation shall show parts breakdown by associated military part number.



Source: https://assist.dla.mil -- Downloaded: 2026-10-19T14:25Z
Check the source to verify that this is the current version before use.

TABLE IV. Configuration.

Code	Configuration	
A 1/	Type 1 Complete assembly with front lens plate	See fig 1
B 1/	" Complete assembly without front lens plate	
C	" Plug-in switch/indicator assy with front lens plate	See fig 3
D	" Plug-in switch/indicator assy without front lens plate	
E	" Housing only	
F 1/	Type 2 Complete assembly with front lens plate	See fig 2
G 1/	" Complete assembly without front lens plate	
H	" Plug-in switch/indicator assy with front lens plate	See fig 3
J	" Plug-in switch/indicator assy without front lens plate	
K	" Housing only	
M 1/	Type 3 Complete assembly with front lens plate	See fig 1
N 1/	" Complete assembly without front lens plate	
P 2/	" Plug-in switch/indicator assy with front lens plate	See fig 3
Q 2/	" Plug-in switch/indicator assy without front lens plate	
R	" Housing only	

1/ For other than direct government stocking, acquisition of assembled units may be made with the code letters indicated. These part numbers shall not appear on the items, but may be marked on shipping containers. The individual part numbers for plug-in switch indicator assembly and housing shall be marked as required.

2/ Not for use in MIL-S-24317/8 multistation switch housings.

TABLE V. Contact action. 1/

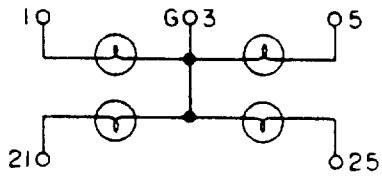
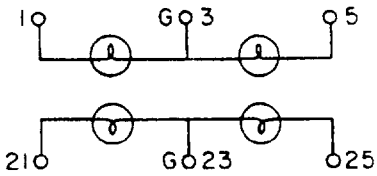
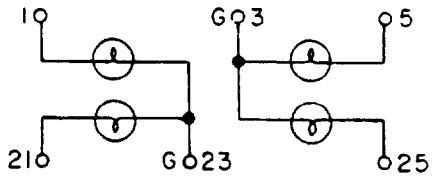
Code	Contact action
A	2PDT Momentary (high impact shock resistant)
B	4PDT Momentary (high impact shock resistant)
C 2/	2PDT Alternate
D 2/	4PDT Alternate
E	Indicator function only (high impact shock resistant)

1/ This code is not required when specifying housing portion of switch only.

2/ Not available with type 3 switches.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T14:25Z
Check the source to verify that this is the current version before use.

TABLE VI. Lens style, lamp circuit and color 1/

Code 2/	Lens style 3/	Lamp schematic
C(X)	1	
D(XX)	1 / 2	
E(XX)	1 2	
F(XXX)	1 2 / 3	
H(XXX)	1 / 2 3	
J(XXXX)	1 2 / 3 4	
K(X)	1	
L(XX)	1 / 2	
M(XXX)	1 2 / 3	
N(XXX)	1 / 2 3	
P(XXXX)	1 2 / 3 4	
S(XX)	1 2	

1/ This code is not required when specifying housing portion of switch only.

2/ Codes are shown for units without color filters When color filters are to be included, add the color symbols of table II in place of (X), (XX), (XXX) or (XXXX) in proper sequence. For direct government stocking, do not use color filter codes, color filters should be acquired separately (see MIL-S-22885/82). Part numbers with color filter codes shall not appear on the items, but may be marked on shipping containers.

3/ Numbers indicate color symbol sequence.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T14:25Z
Check the source to verify that this is the current version before use.

TABLE VII Supersession data 1/

Superseded military part number M22885/80-	Superseding military part number M22885/80-
19 (X)	CAC (X)
20 (XX)	CAL (XX)
21 (XX)	CAS (XX)
22 (X)	CBC (X)
23 (XX)	CBL (XX)
24 (XX)	CBS (XX)
25 (X)	CEC (X)
26 (XX)	CEL (XX)
27 (XX)	CES (XX)
28 (X)	HAC (X)
29 (XX)	HAL (XX)
30 (XX)	HAS (XX)
31 (X)	HBC (X)
32 (XX)	HBL (XX)
33 (XX)	HBS (XX)
34 (X)	HEC (X)
35 (XX)	HEL (XX)
36 (XX)	HES (XX)

1/ Part numbers are shown for units without color filters. When color filters are included, the color symbols of table II are added in place of (X) or (XX) in the sequence shown in table VI.

TABLE VIII. Inactive part numbers. 1/

Part number 2/ M22885/80-	Configuration (see table IV)	Contact action (see table V)	Lens style and lamp circuit (see table VI)
01 (X)	A	A	C
02 (XX)	A	A	L
03 (XX)	A	A	S
04 (X)	A	B	C
05 (XX)	A	B	L
06 (XX)	A	B	S
07 (X)	A	E	C
08 (XX)	A	E	L
09 (XX)	A	E	S
10 (X)	F	A	C
11 (XX)	F	A	L
12 (XX)	F	A	S
13 (X)	F	B	C
14 (XX)	F	B	L
15 (XX)	F	B	S
16 (X)	F	E	C
17 (XX)	F	E	L
18 (XX)	F	E	S

1/ These part numbers are inactive for new design. They cover switches with black painted housing bezel.

2/ Part numbers are shown for units without color filters. When color filters are included, the color symbols of table II are added in place of (X) or (XX) in the sequence shown in table VI.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T14:25Z
Check the source to verify that this is the current version before use.

Custodians
Army - ER
Navy - EC
Air Force - 85

Review activities:
Army - AV, MI
Air Force - 17, 99
DLA - ES
Defense Nuclear Agency - DS

User activities
Army - AR
Navy - MC, SH, AS, OS
Air Force - 19

Agent
DLA - FS

Preparing activity
Navy - EC

(Project 5930-1245-1)

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T14:25Z
Check the source to verify that this is the current version before use.

INSTRUCTIONS In a continuing effort to make our standardization documents better, the DoD provides this form for use in submitting comments and suggestions for improvements All users of military standardization documents are invited to provide suggestions This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)* and mailed In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, was too rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the problems Enter in block 6 any remarks not related to a specific paragraph of the document If block 7 is filled out, an acknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being considered

NOTE This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements

(Fold along this line)

(Fold along this line)

DEPARTMENT OF THE NAVY
Naval Electronic Systems Command
Washington, DC 20363

NO POSTAGE NECESSARY IF MAILED IN THE UNITED STATE

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

BUSINESS REPLY MAIL
FIRST CLASS PERMIT NO 12503 WASHINGTON D C

POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE NAVY

COMMANDER
NAVAL ELECTRONIC SYSTEMS COMMAND
DEFENSE STANDARDIZATION PROGRAM BRANCH
DEPARTMENT OF THE NAVY
WASHINGTON, DC 20363
ATTN: ELEX 8111

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T14:25Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1 DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-S-22885/80E	SWITCH, PUSH BUTTON, ILLUMINATED, 4-LAMP, SOLID MOUNT, RFI SHIELDED, MOISTURE SEALED, HIGH IMPACT SHOCK RESISTAN

3a. NAME OF SUBMITTING ORGANIZATION	4 TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)* _____

5 PROBLEM AREAS

a. Paragraph Number and Wording

b Recommended Wording

c Reason/Rationale for Recommendation

6 REMARKS

7a NAME OF SUBMITTER *(Last, First, MI)* – Optional	b WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c MAILING ADDRESS *(Street City State, ZIP Code)* – Optional	8 DATE OF SUBMISSION *(YYMMDD)*

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T14:25Z
Check the source to verify that this is the current version before use.